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Requirement Analysis and Specification Document

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Contents

Table of Contents . . . 3
List of Figures . . . 4
List of Tables . . . 4

1 Introduction **6**
1.1 Purpose . . . 6
1.1.1 Goals . . . 6
1.2 Scope . . . 7
1.2.1 World Phenomena . . . 7
1.2.2 Shared Phenomena . . . 7
1.3 Definitions, Acronyms, and Abbreviations . . . 8
1.3.1 Acronyms . . . 8
1.4 Revision History . . . 8
1.5 Reference Documents . . . 8
1.6 Document Structure . . . 8

2 Overall Description **9**
2.1 Product Overview . . . 9
2.1.1 Scenarios . . . 9
2.1.2 Domain Class Diagram . . . 12
2.1.3 State Diagrams . . . 13
2.2 Product Functions . . . 18
2.3 User Characteristics . . . 21
2.3.1 Registered User . . . 21
2.3.2 Unregistered User . . . 21
2.4 Assumptions, dependencies and constraints . . . 21
2.4.1 Assumptions . . . 21
2.4.2 Dependencies . . . 22
2.4.3 Constraints . . . 22

3 Specific Requirements **23**
3.1 External Interface Requirements . . . 23
3.2 Functional Requirements . . . 23
3.2.1 Use case diagrams . . . 24
3.2.2 Use cases . . . 24
3.2.3 Sequence diagrams . . . 27

4 Formal Analysis Using Alloy **28**

5 Effort Spent **29**

List of Figures

2.1 Domain Class Diagram 12
2.2 Path Creation State Diagram 13
2.3 Activity Recording State Diagram 14
2.4 Meteorological Data Retrieval State Diagram 15
2.5 Path Research State Diagram 16
2.6 User Lifecycle State Diagram 17

List of Tables

3.2 Use Case: [User Registration] 24
3.4 Use Case: [Creation of a Manual Path] 25
3.6 Use Case: [Publication of a Path] 26
3.8 Use Case: [Search for a Path] 27
3.10 Use Case: [Nome Use Case] 27

5.1 Hours of work spent by each author redacting the RASD document 29

Chapter 1

Introduction

1.1 Purpose

The purpose of the *Best Bike Paths (BBP)* system is to promote cycling as a sustainable, safe, and conscious means of transportation by providing cyclists with a digital platform for collecting, accessing, and sharing information about cycling routes and activities.
The system addresses the need for reliable and up-to-date data on bike paths, allowing users to track their performance while contributing to the improvement of shared information.
Through active community participation, BBP aims to identify and maintain the safest and most efficient routes, fostering a more informed and widespread cycling mobility.
Ultimately, BBP seeks to enhance the overall cycling experience by integrating multiple functionalities and creating a collaborative, continuously updated information network driven by user contributions.

1.1.1 Goals

G1. **Support cyclists in tracking their activities:** Enable users to record their cycling routes and access detailed statistics such as total distance, average speed, and other performance indicators, enriched, when available, by meteorological data from external services (weather conditions, temperature and wind speed).

G2. **Allow cyclists to contribute information about bike paths:** Provide two modes for entering information related to bike paths:

- **Manual mode:** the user manually inputs the path data and its condition.
- **Automatic mode:** the system automatically collects data via GPS, accelerometer, and gyroscope during the cycling activity, requiring user confirmation before publication.

G3. **Enable the publication of recorded path information:** Allow users to make their recorded path information public, contributing to a shared and valuable database for the entire cycling community.

G4. **Enable route search and visualization:** Allow users to search for and display on a map one or more cycling routes connecting a chosen origin and destination, ranking them based on overall condition and effectiveness in reaching the goal.

G5. **Maintain up-to-date and reliable path information:** Ensure continuous updating of path data through the aggregation and analysis of user feedback, taking into account data freshness, consistency among reports, and user confirmations.

1.2 Scope

The main goal of the *Best Bike Paths (BBP)* system is to provide cyclists with a mobile application that allows them to record their activities and share information about bike routes, while also fostering the creation of an active and collaborative community. The application enables registered users to save their rides, view detailed statistics such as total distance covered or average speed, and contribute to updating the path inventory by reporting the condition of routes or the presence of potential obstacles. The information entered by users can be made public and is then aggregated by the system, which combines data from multiple sources to offer a comprehensive and up-to-date representation of the paths' conditions. Unregistered users can still take advantage of the consultation features by searching for routes between two points and viewing the available options on a map, ranked according to a score that considers both the quality and effectiveness of the routes. The system also interacts with an external meteorological service, which provides weather data to enrich the information related to recorded activities.

1.2.1 World Phenomena

WP1. Cyclists travel through real-world routes by following predefined paths.

WP2. Cyclists perceive the quality of the route.

WP3. Cyclists search for routes between two real-world points.

WP4. The physical condition of routes can vary over time.

WP5. Mobile devices can detect data about the cyclist in the real world.

WP6. Real-world weather conditions are detected by meteorological services.

WP7. There are bike lanes and roads where speed limits or the absence of cars make them suitable for bicycles.

1.2.2 Shared Phenomena

User-System

SP1. Users can register within the application.

SP2. The user can record and store their cycling activities.

SP3. The user can choose whether to publish route-related data.

SP4. The system can provide the user with statistics about routes.

SP5. The user can manually enter information about routes.

SP6. The system can automatically record information, and the user must confirm it.

SP7. The user searches for routes based on a starting point and an ending point.

System-Device

SP8. The system receives data from mobile devices to track routes and detect obstacles.

SP9. The system sends requests to an external service and receives weather data.

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1.3 Definitions, Acronyms, and Abbreviations

1.3.1 Acronyms

- **BBP:** Best Bike Paths
- **DB:** DataBase

1.4 Revision History

1.5 Reference Documents

1.6 Document Structure

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Chapter 2

Overall Description

2.1 Product Overview

2.1.1 Scenarios

A new user subscribes to the platform

1. An athlete lands on the website's landing page for the first time. After browsing the home screen and viewing the services offered by the app, they decide to register.
2. The user clicks the "Sign Up" button, which redirects them to the registration page.
3. At this point, the user must decide whether to register using their own credentials (email and password) or to use an OAuth authentication method, logging in with an account from another service.
4. After filling in the required fields, the user is taken to the second screen of the registration flow, where they are asked to provide additional information needed to calculate certain performance metrics, such as weight, height, age, etc.
5. Finally, the user is successfully registered and is redirected to the application's main screen.

A user records an activity

1. When a user wants to record an activity, they must go to the "Record Trip" panel.
2. From there, in the "Automatic" section, they simply need to press the play button: the app will start recording GPS data and storing the route and detected potholes.
3. Once the route is completed, pressing the stop button will end the recording.
4. At this point, the user will be asked to verify whether the potholes detected by the application are actually present or are false positives, and to confirm the condition of the route.
5. Finally, a prompt will ask the user whether they want to publish the route or keep it private.

A user creates a path manually

1. The user logs into the platform using their credentials.
2. Once authenticated, the user accesses the "My Paths" section of their profile.
3. The user selects the "Add Path" option to start creating a new route.

4. The system opens a dedicated page for creating the path.
5. The user enters the required information, including:
    - the name of the path,
    - the streets that compose the path,
    - the condition of each segment,
    - any obstacles or issues present along the route.
6. After completing the input, the user confirms the creation of the path.
7. The system saves the data and displays a confirmation message.
8. The system also offers the user the option to make the path public, so it can be shared with the community.
9. The newly created path is displayed in the "My Paths" section of the user's profile.

A user searches for a path

1. The user can choose to:
    - log in with their account, or
    - use the platform without authentication.
2. From the home page, the user enters the starting point and the desired destination in the search bar.
3. The system receives the request and queries its database of published paths.
4. The system processes the available options and calculates a score for each path based on:
    - the condition of the segments,
    - the freshness of the information,
    - the effectiveness of the route in connecting the selected points.
5. The system displays the results on the map, ordered in descending order of score.
6. The user reviews the results and can select a specific path to view more details.
7. The system displays detailed information for the selected path, such as:
    - total distance,
    - estimated travel time,
    - overall condition of the route.

User publishes a path

1. The user accesses the BBP system and opens the "My Routes" section.
2. The system displays the list of paths associated with the user, indicating for each one, among other properties, its status (private or public).
3. The user selects a private route and, using the appropriate button, requests its publication.
4. Upon receiving the publication request, the system asks the user to confirm additional information related to the route:

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- In the case of manually created routes, the user can review and, if necessary, modify the additional information previously entered during creation.
- In the case of automatically generated routes, the user must confirm or correct the acquired information (if not already done) and may add further details manually.

5. Once the review is complete, the user confirms the publication of the route.
6. The system verifies the data consistency, assigns a unique identifier to the route, and makes it available to the community within the search and browsing functionality.
7. The system notifies the user of the successful publication and updates the route status from private to public.

User modifies the additional information related to a route

1. The user accesses the Best Bike Paths (BBP) platform and opens the “My Routes” section.
2. The system displays the list of routes associated with the user, indicating for each one, among other properties, its status (private or public).
3. The user selects a route and opens the section dedicated to additional information.
4. The system displays the current information, including:
   - The condition of the roads that make up the route;
   - The presence of obstacles or damaged sections;
   - The origin of each piece of data (manually entered or automatically acquired).
5. The user can modify, correct, or add new information:
   - If the route was created manually, the user can freely update existing data.
   - If the route was created automatically, the user can review the information acquired by the system and/or add new data manually; in any case, they must confirm or correct the automatically acquired information before publication.
6. The system saves the changes and updates the database.

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2.1.2 Domain Class Diagram

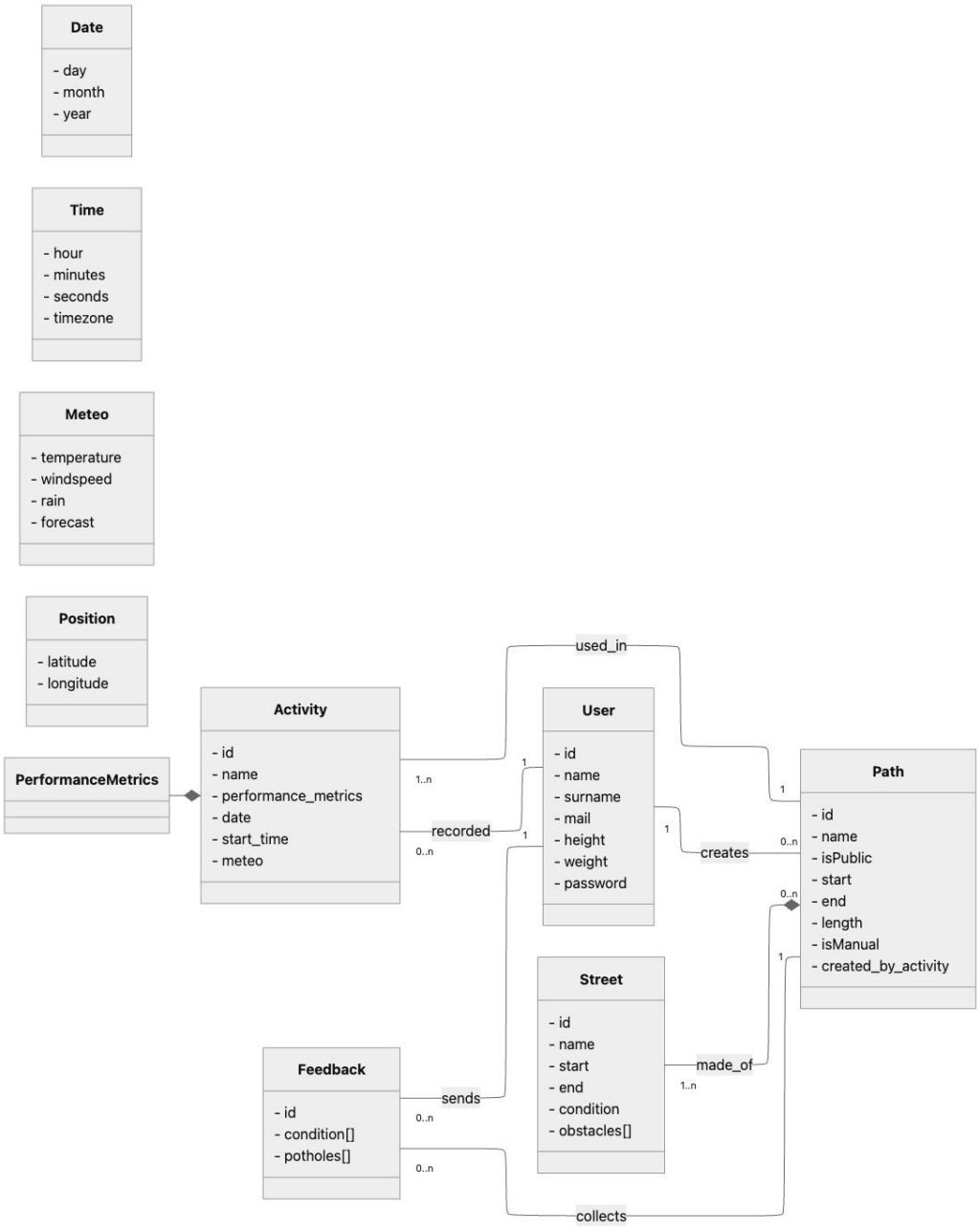


Figure 2.1: Domain Class Diagram

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2.1.3 State Diagrams

Path Creation

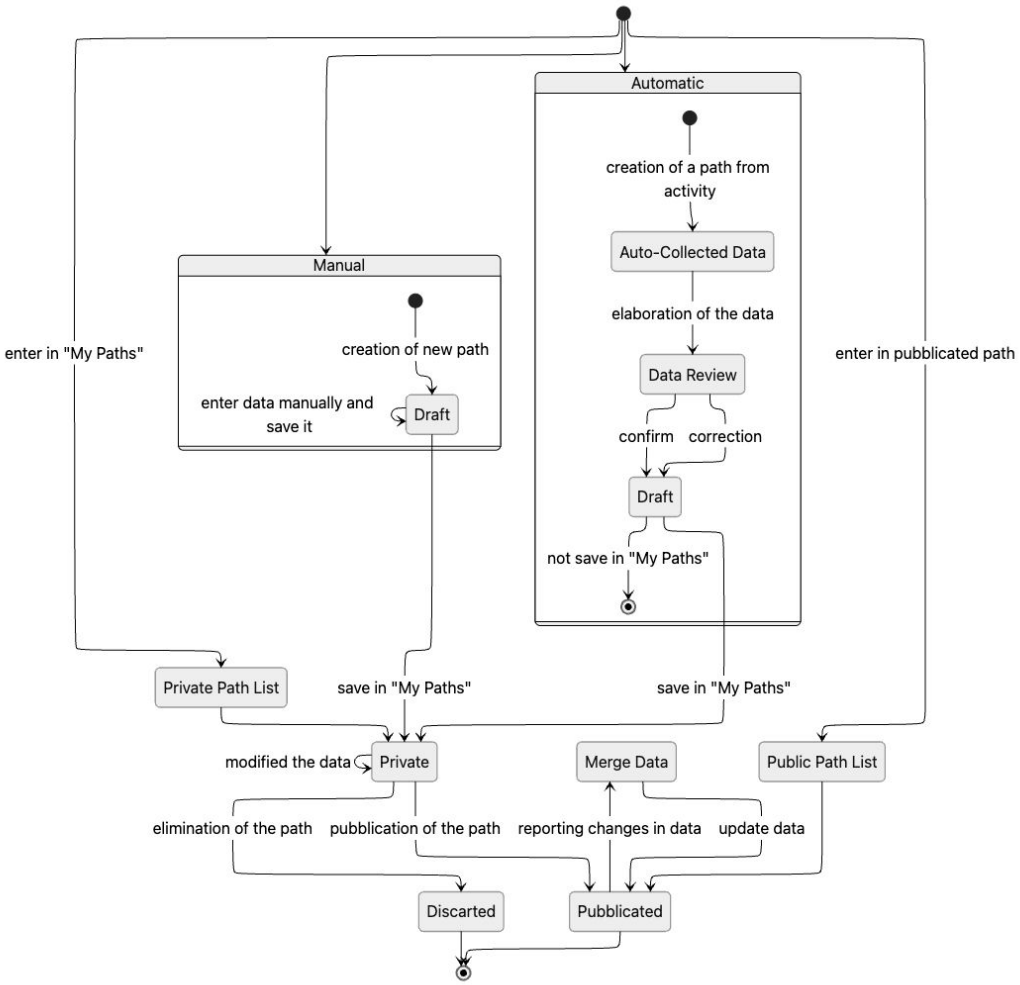


Figure 2.2: Path Creation State Diagram

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Activity Recording

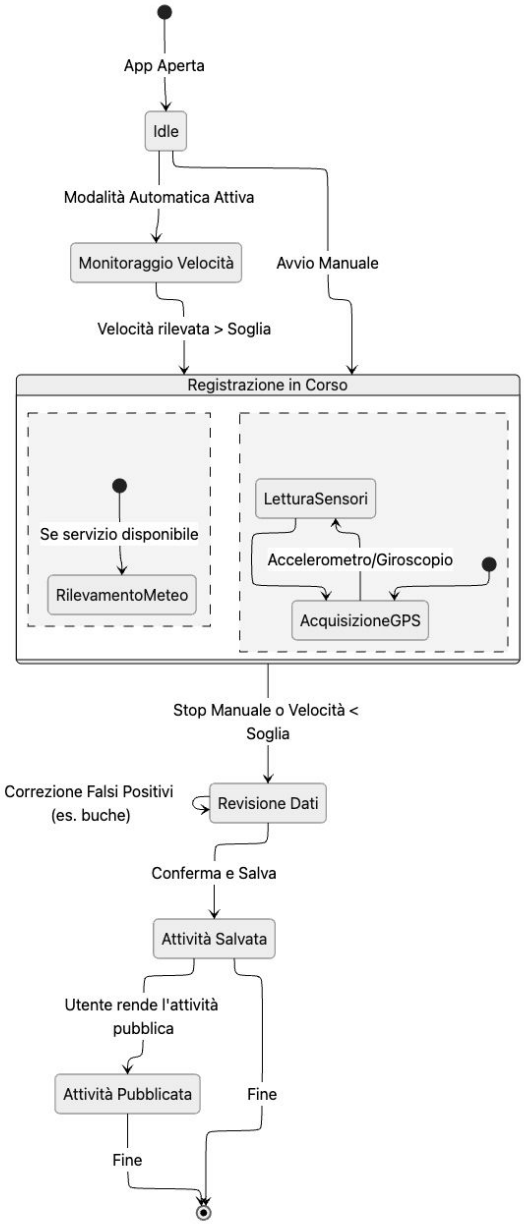


Figure 2.3: Activity Recording State Diagram

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Meteorological Data Retrieval

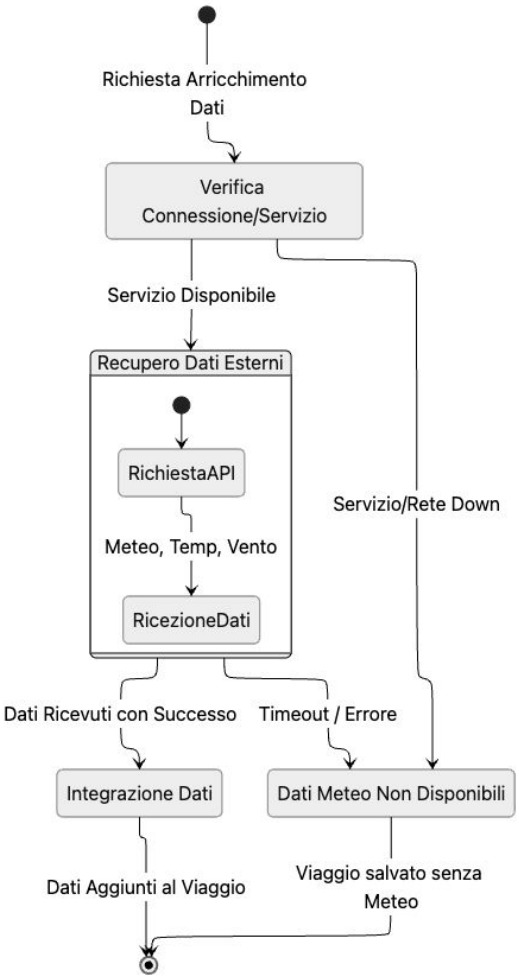


Figure 2.4: Meteorological Data Retrieval State Diagram

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Path Research

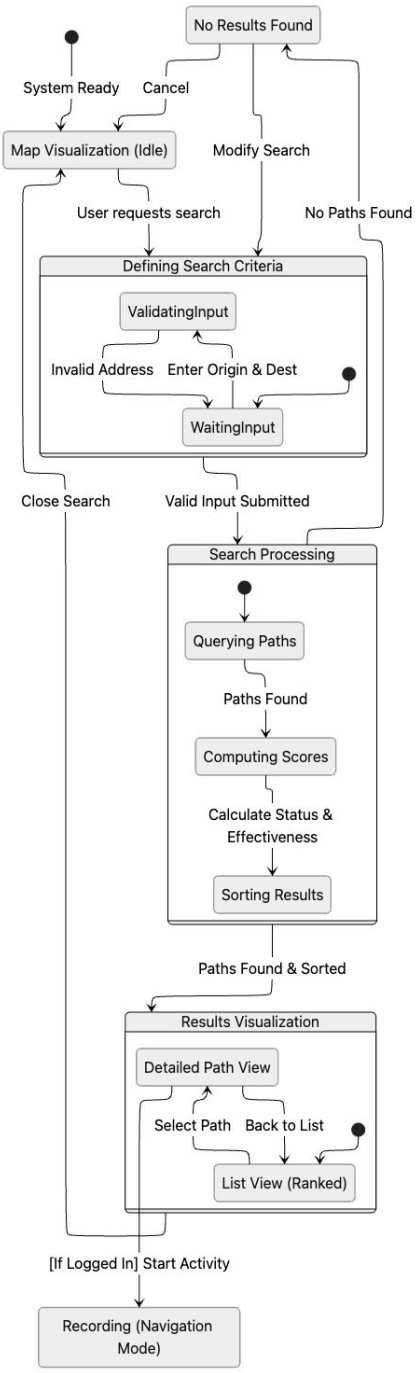


Figure 2.5: Path Research State Diagram

This state machine describes the logic flow of the "Path Search" functionality, accessible to both Guest and Registered users.

The process begins in the **Map Visualization** state. When a user initiates a search, the system transitions to **Defining Search Criteria**, waiting for valid Origin and Destination inputs.

Upon submission, the system enters the **Search Processing** state. This is a critical system-side phase where three operations occur sequentially:

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1. **Querying Paths:** The system retrieves all geometrically valid paths between the two points.
2. **Computing Scores:** The system calculates a specific score for each path, aggregating its maintenance status and its effectiveness in reaching the destination.
3. **Sorting Results:** Paths are ranked based on the computed score.

If valid paths exist, the system transitions to **Results Visualization**. Here, the user can toggle between a ranked list and a **Detailed Path View** (displaying specific obstacles, weather, and stats).

From the detailed view, a specific transition to **Recording (Navigation Mode)** is available, but guarded by the condition `[If Logged In]`, as only registered users can proceed to track an activity on the selected path.

User Lifecycle

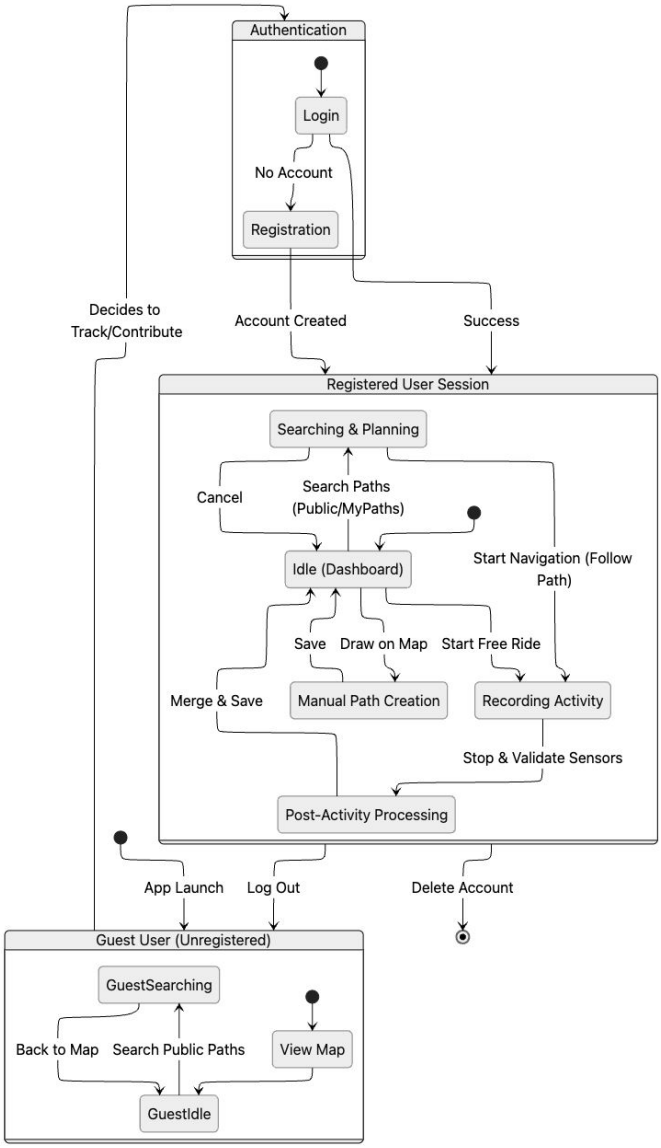


Figure 2.6: User Lifecycle State Diagram

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This state machine diagram describes the user lifecycle within the BBP system, distinguishing between two main macro-states: **Guest User**, unauthenticated, and **Registered User Session**, authenticated.

The lifecycle begins in the **Guest User** state, where functionalities are restricted to read-only operations (represented by the **Guest Searching** state). In this phase, the user can browse the public path inventory but cannot contribute data. To access active functionalities, a transition through the **Authentication** state (Login or Registration) is required.

Once authenticated, the user enters the composite **Registered User Session** state. Here, the **Idle** state acts as a central hub from which three main operational flows branch out:

1. **Searching & Planning:** The user searches for specific paths. From this state, the user can return to the Dashboard or transition directly to **Recording Activity**, following a selected path.
2. **Recording Activity:** This represents the active cycling phase. It can be triggered via the search flow or directly from the Dashboard, not referring to any existing path.
3. **Manual Path Creation:** The user enters an editor mode to manually trace routes on the map without physical recording.

A critical transition occurs at the end of every recording session: the system moves to the **Post-Activity Processing** state. In this mandatory phase, the user validates sensor-detected data and the system executes the merging logic with the existing inventory before returning the user to the **Idle** state. The lifecycle concludes when the user performs a logout or permanently deletes their account.

2.2 Product Functions

Sign-Up and Login

Sign-Up

The system lets the users register in order to manage and record data associated with each of them. When the user accesses the sign-up page for the first time, he's offered the option to log-in with one of their existing accounts through an OAuth service. This way, his BBP account is linked to another account that he owns, making it easier to remember and access for him. The system also gives the user the possibility of entering his email and a new password and to create an account directly inside BBP. In that case, BBP checks that the password is complex enough and that no other account is associated with the same email. If the checks pass, the system stores the information in the DB and grants the user access to the platform.

Login

If the user already has a BBP account or has previously linked their BBP account to one of the other existing accounts that he has through OAuth, the system grants them access the page. If the user access the page through OAuth, a button redirects them to the external account provider's website, which handles the login process and sends the user back only when he completes the login procedure. If, on the other hand, the user registered directly on BBP, the system first checks whether an account associated with the provided email, and then verifies that the password matches. If thats the case the user is redirected to the website's homepage.

Accessing BBP without accounts

The user can also access BBP without logging in. In this case, the functionalities that are available are more limited. The system lets the user browse between public paths, but all the recording features and the private paths needs the login to work

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Activity Recording

The system allows registered users to record their cycling activities. When the user starts the recording, BBP continuously acquires real-time data from the mobile device, including:

- GPS coordinates used to compute distance, average speed, and elevation;
- timestamps associated with all collected events;
- accelerometer and gyroscope data to detect irregularities on the road surface;
- potential stops inferred from speed variations.

During or after the trip, the system processes the collected data and generates a summary reporting distance, duration, average speed, and detected anomalies. If the user decides to save the activity, BBP stores it in the database and makes it available for personal statistics.

Automatic Path Creation

When the user enables the automatic mode, BBP reconstructs the bike path followed during the activity. This includes:

- continuous acquisition of GPS coordinates;
- segmentation of the route based on street structure and direction changes;
- detection of potholes, vibrations, or obstacles through accelerometer and gyroscope data;
- estimation of the user's speed to determine whether they are biking.

At the end of the activity, BBP generates a draft path containing:

- a map of the complete route;
- a list of detected street segments;
- automatically identified road conditions and anomalies.

Since the automatically detected data may contain false positives, the user must review, confirm, or correct all generated information before it becomes publishable. After confirmation, the path and related annotations are stored and made available to the community.

Meteorological Data Integration

When available, BBP enriches recorded activities with meteorological information obtained from an external weather service. After an activity is completed, the system sends:

- the activity timestamps;
- the geographical coordinates of the route.

The external service returns weather conditions for the corresponding time and location, including:

- temperature;
- general weather conditions (e.g., sunny, rainy, cloudy);
- wind speed and direction;
- possible weather alerts.

BBP associates the retrieved meteorological data with the recorded activity before storing it. These details are shown in the activity summary and are available for statistical analysis.

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Manual Path Creation

In addition to the automatic creation of bike paths, registered users can also create paths manually by directly entering the required data. During this process, the platform asks the user to specify the name of the path, the streets that compose it, the condition of each segment, and the possible presence of obstacles or issues. Once the information has been confirmed, the system stores the data in its database and allows the user to decide whether to make the path publishable, meaning visible to the entire community. If the path is not made publishable, it remains visible only to the user who created it, within their personal profile.

Path Search by Origin and Destination

Any user, including unregistered ones, can search for a bike path between two points using the search algorithm provided by the platform. The system queries its database of publishable paths and identifies those that connect the specified origin and destination points. The results are displayed on an interactive map and sorted in descending order based on a path score automatically computed by the platform. The user can select one of the displayed paths to view detailed information, such as the total distance, estimated travel time, and overall path conditions.

Path Ranking

The score assigned to each path is determined by considering several factors:

- Path condition, where better-maintained paths receive higher scores;
- Freshness of information, favoring paths with more recent data;
- Path effectiveness, meaning how efficiently the path connects the two selected points.

Entering route condition information

Each route, whether derived from an automatically recorded activity or manually created by the user, is characterized by a set of additional information describing its condition and rideability. In particular, this information concerns:

- The condition of the roads that make up the route;
- The presence of obstacles, such as potholes, surface damage, or dangerous sections.

Additional information is always managed through the same interface, but with different behavior depending on how the route was created:

- **Manual mode** During route creation, the user directly enters information about the condition of the roads and any obstacles.
- **Automatic mode** The system acquires data via the device's sensors (GPS, accelerometer, gyroscope); therefore, the user finds this information already filled in. Each piece of information must be confirmed.

Regardless of the mode, the user can always modify or update the additional information before and after publication, thus contributing to improving the quality of shared data.

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Paths publication

Registered users can choose to publish a route, making it visible and accessible to other users within the platform. In particular, publication adds the route to *BBP*'s public archive, making it available to all other users, including non-registered ones, through the route search and browsing functionality. Only published routes can be viewed by others and contribute to enhancing the quality of shared community data. Publication always concerns the complete route, including all additional information such as road conditions and the presence of any obstacles.

Merging information from different users

When multiple users provide information regarding the same route or overlapping sections, the BBP system automatically merges the received data concerning road conditions and the presence of obstacles. The merging process takes into account two main factors:

- The freshness of the information, meaning how recently it was provided;
- The number of confirmations received from different users for the same condition or status.

In the case of conflicting reports, the system applies a weighted evaluation mechanism that prioritizes the most recent and most frequently confirmed information.

2.3 User Characteristics

2.3.1 Registered User

A registered user has a personal profile containing the information provided during registration, the saved paths, and the cycling activities they have performed. They can save their rides, view detailed statistics (such as distance, average speed, and total time), and contribute to updating the path inventory by reporting the condition of the routes or the presence of obstacles. The registered user can also create new paths, either manually by entering data directly or automatically during an activity, using the sensors of their mobile device. After creation, the user can make these paths publishable, allowing the community to access and view them. Finally, the user can search for bike paths between two points and view them on the map, ordered according to the score assigned by the system.

2.3.2 Unregistered User

An unregistered user can access a limited version of the application, which only allows the consultation of publishable paths available on the platform. They can search for paths between two points, display them on the map, and view basic information such as the name, score, and general condition of each path, but they cannot perform any additional actions on them.

2.4 Assumptions, dependencies and constraints

2.4.1 Assumptions

A1. **Path Optimality Definition**: A path is considered "better" than another if it minimizes the travel distance or if its start/end points are closer to the user's desired origin/destination.

A2. **Validation Constraint**: While user confirmation can happen anytime for private activities, automatically detected data (sensors) must be validated by the user before being used for public updates.

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A3. **Recording Implies Path Data**: It is assumed that recording a sporting activity implicitly involves collecting path data (GPS/Status) for potential path creation.

A4. **Contribution via Merging**: Once a path is public, its core geometry cannot be overwritten; updates happen through feedback/reports that are merged by the system.

A5. **Manual Creator Competence**: It is assumed that when a user creates a path manually, they have physically traversed it and possess knowledge of its current status.

A6. **Path Uniqueness**: The system assumes that no two identical public paths (same geometry) can exist in the database.

A7. **Manual Duplicate Avoidance**: Automatic duplicate detection relies on GPS tracks. For manual creation, it is assumed the user checks the map for duplicates before creating a new path.

2.4.2 Dependencies

D1. **External weather service**: The system relies on the availability and accuracy of GPS data from users' devices to record cycling activities and create paths.

D2. **Mapping service**: Offers geographic data and map layers used to visualize routes, match GPS tracks to road segments and retrieve details about paths, intersections, and terrains.

D3. **Storage database**: Stores all platform data, including user routes, additional information (road conditions, obstacles), user profiles and publication metadata. Ensures persistence, consistency, and efficient retrieval.

D4. **Authentication services**: Manages user identity verification, login processes and secure access to personal data and features.

D5. **Icon service**: Provides a library of graphical icons used across the interface for consistent and intuitive representation of road conditions, obstacles, route statuses, and other UI elements.

2.4.3 Constraints

C1. **Privacy regulations**: The system must comply with relevant data protection laws and regulations (e.g., GDPR) to ensure user data privacy and security.

C2. **Automatic recording Accuracy limits**: The automatic activity recording feature will operate within the technical accuracy limits of mobile device sensors (GPS, accelerometer, gyroscope). The system must account for possible measurement noise, positioning errors and sensor variability across devices.

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Chapter 3

Specific Requirements

3.1 External Interface Requirements

3.2 Functional Requirements

Sign-Up and Login

1. The platform allows users to create an account by providing the required personal information.
2. The platform allows registered users to log in and access the system.
3. The platform allows registered users to recover the access credentials.
4. The platform allows guest users (non-registered users) to access only the path search functionality of the system.

Path search from origin to destination

1. The platform allow users to search for cycling paths based on a starting point and an ending point.
2. The platform allow users to filter the search results based on the path parameters.

Path Ranking

1. The platform is able to rank the available paths based on the path data, including the condition of each segment, the freshness of the information, and the effectiveness of the path in connecting the selected points.

Entering path condition information

1. The platform is able to detect when a registered user follows a path that is already available to the public and associate the activity with that path.
2. The platform allow registered users to update the condition information of a path, even without performing an activity on that path.

3.2.1 Use case diagrams

3.2.2 Use cases

[UC1] User Registration

Name	**User Registration**
Actors	User, BBP App, Auth service
Entry Condition	User open BBP application without being authenticated
Event Flow	(a) The system displays the welcome screen with options: “Log In”, “Sign Up”, “Continue as Guest” (b) The user selects “Sign Up” (c) The system shows two options: Sign up with an external OAuth provider or create an account with email and password Main Flow (Standard Email Registration) : (d1) The user chooses email registration (e1) The user enter an email and a password (f1) The system verifies that the password is complex enough and that no existing account is associated with the given email (g1) The system stores the new user information in the DB (h1) The system grants the user access to the platform as a registered user Alternative Flow (OAuth Registration): (d2) The user selects an external authentication provider (e2) The Authentication Service validates the user’s identity (f2) The system creates a new account linked to the OAuth identity (g2) The system grants the user access to the platform as a registered user
Exit Condition	The user accesses the system with a Registered Session
Exception	Password Too Simple: the system shows an error message and asks the user to enter a stronger password. Email Already Used: the system alerts the user and suggests entering a different email or logging in with the existing account. OAuth Authentication Failed: the system displays an error and returns to the authentication options.

Table 3.2: Use Case: [User Registration]

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[UC4] Creation of a Manual Path

Name	**Creation of a Manual Path**
Actors	User, BBP App, BBP Server
Entry Condition	The user is authenticated and navigates to the “My Paths” section
Event Flow	(a) The user selects “Create New Path” (b) The platform presents a form requesting all the necessary information of the new path (c) The user enters the data and submit the form (d) The BBP server validates the submitted data (e) The system checks whether a similar path already exists in the public database (f) The platform create a new private path associated with the user (g) The BBP platform confirms creation and displays the new path in the “My Paths” section
Exit Condition	A new private path is successfully created and visible within the user’s “My Paths” section
Exception	Duplicate Public Path Found: the system informs the user that a very similar path already exists in the public database and suggests viewing the public version instead. The new path is not created Invalid or Incomplete Data: the system displays an error and asks the user to correct the form.

Table 3.4: Use Case: [Creation of a Manual Path]

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[UC5] Publication of a Path

Name	**Publication of a Path**
Actors	User, BBP App, BBP Server
Entry Condition	The user is logged in and navigates to the “My Paths” section. The user has at least one private
Event Flow	(a) The user selecta a specific path to view its details (b) The system shows the path information and available actions, including “Publish Path” (c) The user click on the “Publish Path” (d) The Platform validates the request (e) The platform makes the path public and visible to all the users
Exit Condition	The path is stored in the system with status Public, and it becomes visible to all platform users
Exception	Path Already Public: the system informs the user that the selected path is already public Unauthorized User: the user attempts to publish a path they do not own and the system denies the action

Table 3.6: Use Case: [Publication of a Path]

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[UC7] Search for a Path

Name	**Search for a Path**
Actors	User, BBP App, BBP Server
Entry Condition	The user is in the "Search" section of the application (authentication not required)
Event Flow	(a) The user enter the starting and the destination point of the path, that he wants to search (b) The user may optionally apply filters (c) The system validates the user inputs (d) The platform queries the database for all public paths that satisfy the user's search parameters and the applied filters (f) The system computes the path score for each candidate route based on: path condition, freshness of a data and effectiveness in connecting origin and destination (g) The system orders the results in descending order according to the score (h) The platform displays the matching paths on an interactive map and provides a list view (i) The user may select a specific path to see its details
Exit Condition	The BBP platform displays all the search results, sorted by path score, and shows them on the map
Exception	No Result Found: the system displays the message: "No public paths match your search criteria."

Table 3.8: Use Case: [Search for a Path]

[UC1] User Registration

Name	**[Inserisci Nome Use Case]**
Actors	[Inserisci lista Attori]
Entry Condition	[Inserisci Condizione di Ingresso]
Event Flow	(a) [Primo passo del flusso...] (b) [Secondo passo del flusso...] (c) [Terzo passo del flusso...] (d) [Quarto passo del flusso...]
Exit Condition	[Inserisci Condizione di Uscita]
Exception	[Inserisci eventuali Eccezioni]

Table 3.10: Use Case: [Nome Use Case]

3.2.3 Sequence diagrams

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Chapter 4

Formal Analysis Using Alloy

Organize this section according to the rules defined in the project description.

Chapter 5

Effort Spent

Chapter	Date	Alessio Pesavento	Alberto Spezzapria	Luca Suzani
Introduction	27/10/2025	1h	1h	1h
	07/11/2025	2h	2h	2h
	08/11/2025	-	1h	-
	09/11/2025	1h	-	-
	10/11/2025	-	-	1h
	Total	**4h**	**4h**	**4h**
Overall Description	11/11/2025	1.5h	1.5h	1.5h
	12/11/2025	-	1h	-
	13/11/2025	1.5h	1h	2h
	15/11/2025	1h	-	-
	29/11/2025	-	-	1h
	11/12/2025	2.5h	-	-
	Total	**6.5h**	**3.5h**	**4.5h**
Product Functions	22/11/2025	2h	2h	2h
	25/11/2025	1h	1h	1h
	09/12/2025	-	0.5h	1.5h
	11/12/2025	-	-	1h
	Total	**3h**	**3h**	**3h**
Requirements	10/12/2025	2h	-	-
	Total	**2h**	**0h**	**0h**

Table 5.1: Hours of work spent by each author redacting the RASD document